

HARISH NAMASIVAYAM MUTHUSWAMY

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Data & Analytics Professional with 2+ years of experience in data engineering, analytics, business intelligence, and machine learning. Proficient in SQL, Python, ETL/ELT pipeline development, and data warehousing, with hands-on experience building production pipelines, predictive models, and Power BI/Tableau dashboards in financial services and manufacturing environments. Adept at translating complex datasets into actionable insights that drive measurable business outcomes and support data-driven decision-making across cross-functional teams.

TECHNICAL SKILLS

Programming & Query: SQL (CTEs, window functions, stored procedures, query optimization), Python, R

Data Engineering: ETL/ELT Pipelines, PySpark, Airflow, dbt, Delta Lake, Star Schema, Data Warehousing

Analytics & BI: Power BI (DAX, Power Query), Tableau, Grafana, Excel

Machine Learning & Statistics: scikit-learn, Random Forest, Time Series Forecasting, Feature Engineering, A/B Testing

Data Manipulation & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit

Databases & Warehouses: PostgreSQL, MySQL, Microsoft SQL Server, Snowflake, BigQuery, Oracle, SQLite

Cloud & Tools: AWS (S3, EC2, Redshift), GCP, Git, CI/CD, Docker, Jira, Confluence

WORK EXPERIENCE

DATA ANALYTICS & ALGORITHMS INTERN - U-Sense.IT srl, Remote

Mar 2026 - May 2026

- Conducted Exploratory Data Analysis (EDA) across 7 analytical notebooks on 2,000+ automotive sensor records spanning 2 production stages, investigating 15+ data quality issues and surfacing correlation insights to support quality control recommendations.
- Engineered a per-vehicle risk scoring system using 10+ tolerance-margin features to proactively flag high-risk vehicles before Final Assembly, enabling early intervention and reducing costly late-stage rework.
- Built and evaluated 3 predictive models (Linear Regression, Random Forest, Gradient Boosting) using scikit-learn to forecast Final Assembly quality, achieving up to 85% prediction accuracy and enabling proactive defect intervention.
- Designed and deployed a real-time Grafana dashboard tracking 5+ quality KPIs, pass/fail rates, and anomaly alerts across 2 manufacturing stages, driving data-informed decisions for operators and management.

PROJECT ENGINEER - Wipro Technologies Ltd., Chennai, India

Dec 2022 - Jul 2024

- Aggregated and validated data across multiple source systems for a US-based banking client's KYC compliance platform, processing 1M+ customer onboarding records, ensuring regulatory-grade data accuracy through documented pipeline structures.
- Authored and optimized complex SQL queries, stored procedures, and views for data extraction and validation across multiple databases, driving a 40% improvement in ETL performance and reducing processing time.
- Designed Power BI dashboards tracking onboarding KPIs for cross-functional stakeholders, directly contributing to a 15% increase in account activation and a 30% reduction in manual reporting effort.
- Performed root cause analysis on data quality issues across source systems and delivered pipeline-improvement recommendations to cross-functional teams, documenting findings in data dictionaries and lineage documentation.
- Collaborated with QA, business, and compliance teams across 3 Agile/Scrum release cycles, reviewed 20+ pull requests, and resolved 15+ critical defects to improve platform reliability and production readiness.

ANALYTICS INTERN - Wipro Technologies Ltd., Chennai, India

Mar 2022 - May 2022

- Wrote SQL queries to extract, transform, and analyze data from multiple databases, supporting weekly business reports and ad hoc reporting requests across 3+ business units.
- Profiled and validated 50K+ customer records across multiple source systems to identify data inconsistencies and missing values, supporting downstream reporting accuracy for business analytics workflows.
- Assisted in building Excel-based reports and basic dashboards summarizing weekly operational metrics, supporting team analytics workflows and providing visibility into business performance for senior analysts.

TECHNICAL PROJECTS

RetailIQ - Modern Data Platform for Operational Intelligence | *Python, SQL, Airflow, dbt, Snowflake, Power BI*

- Ingested and transformed 100K+ e-commerce transactions using Airflow and dbt into a Snowflake Star Schema with 5+ dimension and fact tables, enabling analytics-ready data delivery in near real-time.
- Designed an interactive Power BI dashboard tracking 8+ KPIs across revenue, customer segments, and product performance, enabling stakeholders to perform drill-down analysis and identify actionable trends.

ChurnShield - Customer Churn Prediction & Growth Analytics | *Python, scikit-learn, Pandas, SQL, Streamlit*

- Engineered 15+ features from 50K+ customer records and built classification models (Random Forest, Gradient Boosting) using scikit-learn, achieving 89% accuracy in identifying at-risk customers.
- Deployed a Streamlit dashboard visualizing churn drivers and risk segments across 5+ KPIs, enabling targeted retention analysis and customer-level risk scoring.

Chicago CTA Transit Analytics Dashboard | *Python, SQL, Pandas, Streamlit, Plotly, SQLite*

- Processed 17,000+ vehicle records and 69,000+ predictions across 14 CTA bus routes into a structured SQLite database, writing 10 analytical SQL queries to explore route delay profiles, on-time performance, and stop-level patterns.
- Designed a Star Schema with fact and dimension tables in SQLite to support drill-down analysis across routes, time windows, and stop locations.
- Built an interactive Streamlit dashboard with 5 KPI cards, delay trend charts, activity heatmaps, and a geographic vehicle map for visual exploration of transit performance.

FinStream - Credit Card Transaction Analytics Pipeline | *Python, SQL, Snowflake, dbt, Pandas*

- Built an end-to-end ETL pipeline processing 100K+ simulated credit card transaction records into a Snowflake analytics layer using Python and dbt, with layered transformations for transaction-level analytics.
- Engineered aggregated features including rolling spend windows, merchant-category groupings, and geo-anomaly indicators to support exploratory fraud-pattern modeling.
- Implemented dbt transformations and Star Schema modeling across fact and dimension tables, with data quality tests across pipeline layers.

FinDocs RAG - Financial Document Q&A System | *Python, LangChain, OpenAI API, FAISS, Streamlit*

- Built a Retrieval-Augmented Generation (RAG) chatbot over 500+ financial compliance and policy documents, using LangChain and a FAISS vector store with sentence-transformer embeddings for semantic retrieval.
- Implemented chunking, overlap, and metadata-aware retrieval strategies to improve answer relevance, and integrated source citations into responses for output traceability.
- Deployed an interactive Streamlit interface for natural-language Q&A over the document corpus, demonstrating end-to-end LLM application development from ingestion through retrieval to response.

EDUCATION

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| • Illinois Institute of Technology, Chicago, Illinois. | May 2026 |
| Master of Applied Science, Data Science. | GPA 3.81 |
| • St. Joseph’s College of Engineering, Chennai, India. | May 2022 |
| Bachelor of Engineering, ECE. | GPA 3.65 |

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate - DP-700
- Associate Cloud Engineer Certification

PUBLICATIONS & ACHIEVEMENTS

IEEE Publication – “Hybrid Transformer and XGBoost Model for Federated IoT Intrusion Detection” | *ICAIC 2026, University of Houston, TX*

- Designed a hybrid Transformer + XGBoost architecture within a federated learning framework with LoRA-based parameter-efficient updates, achieving ~98% detection accuracy while reducing communication overhead by ~90%.

LA Hacks 2025 (UCLA) – Melissa Data Challenge Winner | *FastAPI, React, MongoDB, Google Gemini AI, Melissa APIs*

- Built “Bandaids Maps,” a real-time AI-powered healthcare navigation tool generating personalized Health Readiness Scores and emergency action plans in a 36-hour hackathon build.